Fiscal Year:	FY 2011	**Task Last Updated:**	FY 01/27/2012
PI Name:	Stutte, Gary Ph.D.		
Project Title:	Symbiotic Nodulation in a Reduced Gravity Environment (SyNRGE)		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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PI Organization Type:	NASA CENTER	**Phone:**	321-861-3493
Organization Name:	QinetiQ-North America/NASA Kennedy Space Center		
PI Address 1:	ESC-Team QNA		
PI Address 2:	Space Life Sciences Laboratory		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899	**Congressional District:**	24
Comments:	NOTE: PI had a change of employment from Dynamac Corporation to QinetiQ-North America, effective 1 March, 2011.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	09/01/2010	**End Date:**	02/28/2011
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-135; ISS NOTE: PI's employment changed in March 2011, changing end date to 2/28/2011 (Ed., 1/19/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roberts, Michael (Dynamac Corporation)		
Grant/Contract No.:	NNX10AR09A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	SyNRGE (Symbiotic Nodulation in a Reduced Gravity Environment) will investigate microgravity effects associated with microbe-host interactions and cell-cell communication using a plant-bacteria model system. Seeds from a single cultivar of the model legume species Medicago truncatula (barrel medic) will be germinated on-orbit in the presence of genetically marked strains of nitrogen-fixing bacteria of the species Sinorhizobium meliloti capable of forming a mutualistic endosymbiosis with leguminous plants. Cellular, molecular, and morphological analyses of plant tissue specimens and bacteria fixed at serial time intervals on-orbit will be compared to ground controls to elucidate the effect of microgravity on plant and bacterial physiological responses, mechanisms of cell-cell communication and resource exchange (i.e., carbon and nitrogen), and the extent of symbiotic nodule formation. Symbiotic nitrogen-fixing bacteria and leguminous plants have evolved complex signal exchange mechanisms that enable specific bacterial species to induce specific host plant species to form invasion structures through which the bacterium enters the plant root. Once inside, the bacteria are enclosed within a micro-aerobic cell compartment by the host plant and differentiate into specialized cells that express an oxygen-sensitive enzyme to catalyze the conversion of nitrogen to ammonia for use by the host plant. Although the rhizobia-legume microbe-host relationship is mutually beneficial, rhizobial infections of plants can be deleterious. Rhizobia are similar to Brucella spp. that infect animals (including humans) in that they both form chronic infections of eukaryotic cells within a host-derived membrane compartment and require host-derived factors for survival within the host. Based on recent reports of increased virulence of Salmonella typhimurium, Pseudomonas aeruginosa, and Streptococcus pneumoniae in reduced gravity, differences in the infectious and/or pathogenic responses of rhizobial bacteria or the systemic acquired resistance of the host plant are also likely.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	SyNRGE investigates the role of microgravity on the development of symbiosis between model legume plant species, Medicago truncatula, and its bacterial symbiot, Sinorhizobium meliloti. This plant/bacteria symbiosis allows nitrogen to be fixed from the atmosphere and is responsible for over 20% of the protein consumed in the diet. SyNRGE uses the microgravity environment to investigate mechanisms for biological nitrogen fixation. Understanding of this important plant/bacteria relationship has wide-ranging agricultural and ecological implications.
Task Progress:	The Principal Investigator (Gary Stutte, PhD) and co-Investigator (Mike Roberts, PhD) had a change of employment from Dynamac Corporation, prime contractor of the Life Science Services Contract (LSSC) at Kennedy Space Center, to QinetiQ-North America, the prime contractor for the Engineering Services Contract (ESC) at Kennedy Space Center, effective 1 March, 2011. See "Symbiotic Nodulation in a Reduced Gravity Environment (SyNRGE)--NNX11AL06A" for future reports. Editor: 1/19/2012
Bibliography Type:	Description: (Last Updated: 04/16/2020)